SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO4 ASSESSMENT TOOL 1 – Test Case Design Assignment

Course Code:	CSA10 / CSA1063	**Course Title:**	Software Engineering
CO Assessed:	CO4	**Assessment:**	Assessment Tool 1 – Test Case Design Assignment
Weightage:	20%	**Total Marks:**	25
CO4 Statement:	Implement robust testing, quality assurance, and DevOps practices, emphasizing security, reliability, and ethics		
Student Name:	P. MUNI KARTHIK	**Reg. No.:**	192425061
Date:		**Duration:**	60 minutes

Code of Conduct

I P. MUNI KARTHIK / 192425061 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: P. MUNI KARTHIK

Annexure A – Test Case Design Assignment

Instructions to Students:

Each student is assigned an individual problem statement. Based on the assigned problem statement, **design and document comprehensive test cases** to verify the correctness, functionality, reliability, and usability of the proposed system.

Your test case design should include:

1. Identify the **functional and non-functional requirements** to be tested.
2. Identify the **test scenarios** for the assigned problem statement.
3. Prepare detailed **test cases** covering normal, boundary, invalid, and exceptional conditions.
4. Include **Test Case ID, Test Scenario, Test Steps, Test Data, Expected Result, Actual Result, and Pass/Fail Status**.
5. Apply appropriate **test design techniques** such as Equivalence Partitioning, Boundary Value Analysis, Decision Table, or State Transition Testing wherever applicable.

6. Ensure that the test cases provide adequate **requirement and functional coverage** for the assigned system.

Deliverable: Submit a Test Case Design document containing the identified scenarios, test cases, test data, expected results, and test coverage based on the individual problem statement.

Rubrics for Calculation

Evaluation Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Requirement & Test Scenario Identification(15%)	Clearly identifies all relevant functional/nonfunctional requirements and comprehensive test scenarios	Identifies most requirements and scenarios	Identifies basic requirements and scenarios	Requirements/scenarios are incomplete or unclear
Test Case Design & Completeness(25%)	Comprehensive test cases covering normal, boundary, invalid, and exceptional conditions	Covers most important conditions with minor gaps	Covers basic conditions but misses several important cases	Test cases are very limited or incomplete
Test Data & Expected Results(20%)	Appropriate test data with precise, measurable expected results for every test case	Mostly appropriate test data and expected results	Some test data/results are unclear or incomplete	Test data and expected results are largely missing/incorrect
Application of Test Design Techniques(15%)	Correctly applies multiple techniques such as Equivalence Partitioning, Boundary Value Analysis, Decision Table, and State Transition	Correctly applies at least two relevant techniques	Applies one technique with limited justification	Techniques are not applied or incorrectly applied

Traceability & Test Coverage(15%)	Excellent mapping between requirements, scenarios, and test cases with strong coverage	Good traceability and coverage with minor gaps	Partial traceability and moderate coverage	Poor/no traceability and inadequate coverage
Documentation & Presentation(10%)	Well-structured, clear, professional, consistent, and easy to understand	Well-organized with minor presentation issues	Adequately organized but has some clarity/formatting issues	Poorly organized, unclear, or incomplete documentation

Criteria	**Mark**
Requirement & Test Scenario Identification (15%)	/5
Test Case Design & Completeness (25%)	/4
Test Data & Expected Results (20%)	/4
Application of Test Design Techniques (15%)	/4
Traceability & Test Coverage (15%)	/4
Documentation & Presentation (10%)	/4
TOTAL	/25

DATA WAREHOUSE MANAGEMENT SYSTEM

1. Introduction

A **Data Warehouse Management System (DWMS)** is a centralized system used to collect, transform, integrate, store, and analyze data from multiple sources. It supports business intelligence and decision-making by providing consistent and historical data for reporting and analysis.

The purpose of this test case design is to verify the **correctness, functionality, reliability, performance, security, and usability** of the Data Warehouse Management System. The testing covers data extraction, transformation, validation, loading, storage, querying, reporting, error handling, backup, and access control.

2. Problem Statement

Design and develop a **Data Warehouse Management System** that collects data from multiple sources, performs Extract, Transform, and Load (ETL) operations, stores the processed data in a centralized data warehouse, and provides accurate analytical queries and reports.

The system should ensure that:

- Data is extracted correctly from source systems.
- Invalid and incomplete data is identified.
- Data is transformed into a consistent format.
- Clean data is loaded into the warehouse.
- Duplicate records are handled correctly.
- Fact and dimension tables maintain data integrity.
- Users can retrieve data through analytical queries.
- Reports display accurate information.

3. Objectives of Testing

The main objectives are:

1. Verify that data can be extracted from valid sources.
2. Verify that data transformation rules work correctly.
3. Verify that valid data is loaded into the warehouse.
4. Verify that invalid data is rejected or handled appropriately.
5. Verify data consistency and integrity.
6. Verify duplicate-record handling.
7. Verify relationships between fact and dimension tables.
8. Verify the accuracy of analytical queries.
9. Verify report generation.
10. Verify system performance with large datasets.
11. Verify authentication and authorization.
12. Verify error handling and recovery.
13. Verify backup and restoration functionality.
14. Verify usability of the system.

4. Functional Requirements to be Tested

Requirement ID	Functional Requirement
FR-01	The system shall accept data from supported source systems.
FR-02	The system shall extract source data correctly.
FR-03	The system shall validate incoming data.
FR-04	The system shall identify missing and invalid values.
FR-05	The system shall transform data according to predefined rules.
FR-06	The system shall remove or handle duplicate records.
FR-07	The system shall load valid transformed data into the warehouse.
FR-08	The system shall maintain fact and dimension tables.
FR-09	The system shall maintain referential integrity.
FR-10	The system shall support analytical queries.
FR-11	The system shall generate reports.
FR-12	The system shall maintain historical data.
FR-13	The system shall display ETL job status.
FR-14	The system shall handle ETL failures.
FR-15	The system shall provide user authentication.
FR-16	The system shall provide role-based access control.
FR-17	The system shall support backup and recovery.
FR-18	The system shall maintain logs of important operations.

5. Non-Functional Requirements to be Tested

Requirement ID	Non-Functional Requirement
NFR-01	The system should process data within an acceptable time.

NFR-02	The system should support large datasets.
NFR-03	The system should be reliable during ETL operations.
NFR-04	The system should protect sensitive warehouse information.
NFR-05	The system should provide appropriate error messages.
NFR-06	The system should be easy to use.
NFR-07	The system should maintain data accuracy and consistency.
NFR-08	The system should recover from failures without unnecessary data loss.
NFR-09	The system should provide adequate availability.
NFR-10	The system should maintain audit logs for important operations.

6. Test Scenarios

Scenario ID	Test Scenario
TS-01	Verify user login with valid credentials.
TS-02	Verify login with invalid credentials.
TS-03	Verify role-based access.
TS-04	Verify valid data source connection.
TS-05	Verify invalid data source handling.
TS-06	Verify successful data extraction.
TS-07	Verify missing-value handling.
TS-08	Verify invalid data-type handling.
TS-09	Verify duplicate-record handling.
TS-10	Verify data transformation.
TS-11	Verify data format conversion.
TS-12	Verify successful warehouse loading.
TS-13	Verify failed warehouse loading.
TS-14	Verify fact table integrity.
TS-15	Verify dimension table integrity.

TS-16	Verify primary and foreign key relationships.
TS-17	Verify analytical queries.
TS-18	Verify report generation.
TS-19	Verify historical data storage.
TS-20	Verify ETL job status.
TS-21	Verify ETL error logging.
TS-22	Verify large-volume data processing.
TS-23	Verify concurrent user access.
TS-24	Verify backup functionality.
TS-25	Verify data recovery.
TS-26	Verify session logout.
TS-27	Verify unauthorized access prevention.
TS-28	Verify system response to database failure.
TS-29	Verify audit logging.
TS-30	Verify usability of the system.

---

7. Detailed Test Cases

7.1 Authentication and Access Control

Test Case ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Pass/Fail
TC-001	Valid login	1. Open login page. 2. Enter username and password. 3. Click Login.	Username: admin, Password: Admin@123	User should be successfully logged in and redirected to the dashboard.	To be filled during execution	To be filled
TC-002	Invalid password	1. Open login page. 2. Enter	admin / Wrong@123	Login should fail and an	To be filled	To be filled

		valid username and incorrect password. 3. Click Login.		appropriate error message should be displayed.	during execution	
TC-003	Empty login fields	1. Open login page. 2. Leave username and password empty. 3. Click Login.	Blank username and password	System should prevent login and display validation messages.	To be filled during execution	To be filled
TC-004	Role-based access	1. Login as a normal user. 2. Open administration functions.	User account	Restricted administrative functions should not be accessible.	To be filled during execution	To be filled
TC-005	Logout	1. Login successfully. 2. Click Logout. 3. Try to access dashboard using browser back button.	Valid user account	User should be logged out and protected pages should not be accessible without login.	To be filled during execution	To be filled

7.2 Data Source and Extraction Testing

Test Case ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Pass/Fail
TC-006	Valid data source	1. Configure source connection. 2. Enter valid connection details. 3. Test connection.	Valid database credentials	Connection should be established successfully.	To be filled during execution	To be filled
TC-007	Invalid data source	1. Enter incorrect database credentials. 2. Test connection.	Invalid username/password	Connection should fail and an appropriate error should be displayed.	To be filled during execution	To be filled
TC-008	Successful extraction	1. Select valid source. 2. Start extraction. 3. Monitor job.	Valid source with 1,000 records	All eligible source records should be extracted successfully.	To be filled during execution	To be filled
TC-009	Empty source	1. Connect to an empty	Table with 0 records	System should complete	To be filled	To be filled

		source table. 2. Start extraction.		extraction without corrupting the warehouse and report that no records were available.	during execution	
TC-010	Source connection failure	1. Start extraction. 2. Disconnect or make the source unavailable during extraction.	Unavailable source	Extraction should fail gracefully and the error should be logged.	To be filled during execution	To be filled

7.3 Data Validation and Transformation

Test Case ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Pass/Fail
TC-011	Valid data validation	1. Upload valid dataset. 2. Start validation.	Complete valid records	All valid records should pass validation.	To be filled during execution	To be filled
TC-012	Missing required field	1. Upload dataset containing a missing required	Customer ID = NULL	Record should be rejected or handled according	To be filled	To be filled

		value. 2. Run validation.		to defined business rules.	during execution	
TC-013	Invalid data type	1. Upload dataset. 2. Enter text in a numeric field. 3. Run validation.	Sales Amount = "ABC"	Invalid record should be identified and prevented from being incorrectly loaded.	To be filled during execution	To be filled
TC-014	Duplicate records	1. Upload dataset containing duplicate records. 2. Run ETL.	Two identical customer records	Duplicate records should be detected and handled according to the defined duplicate policy.	To be filled during execution	To be filled
TC-015	Date format conversion	1. Provide dates in supported source format. 2. Run transformation.	21/08/2026	Dates should be converted into the warehouse's standard format.	To be filled during execution	To be filled
TC-016	Negative sales value	1. Upload transaction containing negative sales amount. 2. Run validation.	Sales = -500	System should reject or flag the value according to business rules.	To be filled during execution	To be filled
TC-017	Whitespace handling	1. Upload data containing leading/trailing	" Chennai "	Unnecessary spaces should be	To be filled	To be filled

		spaces. 2. Run transformation.		removed during transformation.	during execution	
TC-018	Null optional field	1. Upload record with optional field missing. 2. Run ETL.	Phone = NULL	Record should be processed if the field is optional.	To be filled during execution	To be filled

---

7.4 Data Loading and Warehouse Integrity

Test Case ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Pass/Fail
TC-019	Successful warehouse load	1. Prepare valid transformed data. 2. Execute load operation. 3. Verify warehouse.	1,000 valid records	All valid records should be loaded correctly.	To be filled during execution	To be filled
TC-020	Invalid record during load	1. Include invalid record. 2. Start loading. 3. Check result.	Invalid customer ID	Invalid record should not corrupt the warehouse and should be logged or rejected.	To be filled during execution	To be filled
TC-021	Fact table loading	1. Execute ETL. 2. Open	Sales transactions	Fact records should contain	To be filled	To be filled

		fact table. 3. Compare source and warehouse values.		correct measures and dimension references.	during execution	
TC-022	Dimension table loading	1. Execute dimension load. 2. Verify dimension records.	Customer/Product data	Dimension records should be inserted or updated correctly.	To be filled during execution	To be filled
TC-023	Referential integrity	1. Load fact data. 2. Check foreign-key references.	Valid Customer IDs	Every fact record should reference a valid dimension record.	To be filled during execution	To be filled
TC-024	Duplicate primary key	1. Attempt to insert duplicate dimension key. 2. Execute load.	Existing Customer ID	Duplicate primary-key records should be prevented or handled according to the defined loading strategy.	To be filled during execution	To be filled

7.5 Query and Reporting Testing

Test Case ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Pass/Fail
TC-025	Simple data query	1. Login. 2. Open query interface. 3. Execute a valid query.	SELECT customer records	Correct records should be displayed.	To be filled during execution	To be filled
TC-026	Aggregate query	1. Execute SUM query. 2. Compare result with source calculation.	Total sales	Calculated total should match the expected value.	To be filled during execution	To be filled
TC-027	Filtered query	1. Apply date/category filter. 2. Execute query.	Year = 2026	Only matching records should be displayed.	To be filled during execution	To be filled
TC-028	Invalid query	1. Enter invalid SQL/query syntax. 2. Execute query.	Invalid query syntax	System should display an understandable error without crashing.	To be filled during execution	To be filled
TC-029	Report generation	1. Select report type. 2. Select required filters. 3. Generate report.	Monthly Sales Report	Report should be generated with accurate data.	To be filled during execution	To be filled
TC-030	Empty report result	1. Select a period with no records.	Future date with no data	System should display an appropriate empty-	To be filled	To be filled

		2. Generate report.		result message instead of an incorrect report.	during execution	

7.6 ETL Job and Error Handling

Test Case ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Pass/Fail
TC-031	ETL job starts	1. Select valid ETL job. 2. Click Start.	Valid ETL configuration	Job status should change from Pending to Running.	To be filled during execution	To be filled
TC-032	ETL successful completion	1. Start valid ETL job. 2. Wait for completion.	Valid dataset	Status should change to Completed and records should be loaded.	To be filled during execution	To be filled
TC-033	ETL failure	1. Start ETL with invalid source. 2. Monitor status.	Invalid source	Job should change to Failed and provide an appropriate error message.	To be filled during execution	To be filled
TC-034	Error logging	1. Generate an ETL error. 2. Open system logs.	Invalid data	Error details should be recorded in the log.	To be filled during execution	To be filled
TC-035	ETL retry	1. Allow ETL job to fail. 2.	Corrected source	ETL should restart and	To be filled	To be filled

		Correct the problem. 3. Retry the job.		complete successfully when the source is valid.	during execution	

7.7 Performance, Backup, Recovery and Security

Test Case ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Pass/Fail
TC-036	Large dataset processing	1. Load a large dataset. 2. Start ETL. 3. Monitor processing.	100,000 records	System should process the dataset without crashing or unacceptable degradation.	To be filled during execution	To be filled
TC-037	Concurrent users	1. Login using multiple user accounts. 2. Execute queries simultaneously.	10 concurrent users	System should remain available and return correct results.	To be filled during execution	To be filled
TC-038	Unauthorized access	1. Login as restricted user. 2. Attempt to access administrative URL/function.	Normal user	Access should be denied.	To be filled during execution	To be filled
TC-039	Database backup	1. Open backup function. 2. Start	Current warehouse	Backup should complete successfully and	To be filled	To be filled

		backup. 3. Verify backup file/status.		contain the required warehouse data.	during execution	
TC-040	Database recovery	1. Restore warehouse from a valid backup. 2. Verify important tables and records.	Previous backup	Warehouse should be restored correctly without unacceptable data loss.	To be filled during execution	To be filled

8. Test Design Techniques

8.1 Equivalence Partitioning

Equivalence Partitioning divides input data into valid and invalid groups. One representative value from each group is tested.

Example: Sales Amount

Partition	Test Data	Expected Result
Valid positive value	5000	Accepted
Zero	0	Accepted/rejected according to business rule
Negative value	-500	Rejected or flagged
Non-numeric value	ABC	Rejected

This reduces the number of test cases while maintaining effective input coverage.

8.2 Boundary Value Analysis

Boundary Value Analysis tests values at the edges of an acceptable range.

Example: Number of Records

Assume the system supports a maximum batch size of **100,000 records**.

Test Case	Input	Expected Result
Below boundary	99,999	Accepted
At boundary	100,000	Accepted
Above boundary	100,001	Rejected or handled according to system limitation
Minimum	1	Accepted
Below minimum	0	Handled according to system rules

8.3 Decision Table Testing

Decision Table Testing is useful when the result depends on multiple conditions.

ETL Loading Decision Table

Condition	Rule 1	Rule 2	Rule 3	Rule 4
Source available	Yes	Yes	No	Yes
Data valid	Yes	No	Yes	No
Required fields present	Yes	Yes	Yes	No
Action	Load data	Reject/flag data	Report source error	Reject/flag data

Expected Results

- If the source is available and data is valid, the data should be loaded.
- If data is invalid, the system should reject or flag the affected records.
- If the source is unavailable, the ETL operation should fail gracefully.
- If required fields are missing, the affected records should not be incorrectly loaded.

8.4 State Transition Testing

State Transition Testing verifies whether the system moves correctly between different states.

ETL Job States

Pending → Running → Completed

or

Pending → Running → Failed → Retry → Running → Completed

Test Case	Current State	Event	Expected Next State
ST-01	Pending	Start ETL	Running
ST-02	Running	ETL successful	Completed
ST-03	Running	ETL error	Failed
ST-04	Failed	Retry	Running
ST-05	Completed	View result	Completed
ST-06	Pending	Cancel job	Cancelled

The system should not allow invalid state transitions, such as marking a job as Completed before it has started.

---

9. Requirement Traceability Matrix

Requirement	Covered Test Cases
FR-01 Data source support	TC-006, TC-007
FR-02 Data extraction	TC-008, TC-009, TC-010
FR-03 Data validation	TC-011, TC-012, TC-013
FR-04 Missing values	TC-012, TC-018
FR-05 Data transformation	TC-015, TC-017
FR-06 Duplicate handling	TC-014, TC-024
FR-07 Warehouse loading	TC-019, TC-020

FR-08 Fact/dimension tables	TC-021, TC-022
FR-09 Data integrity	TC-023, TC-024
FR-10 Analytical queries	TC-025, TC-026, TC-027
FR-11 Reports	TC-029, TC-030
FR-12 Historical data	TC-027, TC-029
FR-13 ETL status	TC-031, TC-032, TC-033
FR-14 ETL failure handling	TC-033, TC-034, TC-035
FR-15 Authentication	TC-001, TC-002, TC-003
FR-16 Authorization	TC-004, TC-038
FR-17 Backup/recovery	TC-039, TC-040
FR-18 Logging	TC-034

10. Test Coverage

The designed test cases provide coverage for the major functional and non-functional areas of the Data Warehouse Management System.

Testing Area	**Number of Test Cases**
Authentication & Access Control	5
Data Source & Extraction	5
Data Validation & Transformation	8
Data Loading & Integrity	6
Query & Reporting	6
ETL & Error Handling	5
Performance, Security & Recovery	5
Total	**40**

Test Design Technique Coverage

Technique	Application
Equivalence Partitioning	Valid/invalid data, numeric values, source data
Boundary Value Analysis	Dataset size, numeric limits, input limits
Decision Table Testing	ETL loading decisions
State Transition Testing	ETL job lifecycle
Requirement-Based Testing	Functional and non-functional requirements

11. Expected Test Execution Summary

The following table can be completed after executing the test cases on the actual system.

Metric	Value
Total Test Cases	40
Executed	To be filled
Passed	To be filled
Failed	To be filled
Blocked	To be filled
Not Executed	To be filled
Pass Percentage	To be calculated

Pass Percentage

Pass Percentage = (Number of Passed Test Cases / Number of Executed Test Cases) × 100

12. Defect Classification

During execution, detected defects can be classified as follows:

Severity	Description	Example
Critical	Prevents the core system from functioning	Warehouse database cannot be accessed

High	Major functionality is unavailable	ETL cannot load any valid records
Medium	Important functionality has limited impact	Incorrect filtering in reports
Low	Minor usability or display issue	Misaligned text or minor UI issue

13. Acceptance Criteria

The Data Warehouse Management System can be considered ready for acceptance when:

1. Valid data is successfully extracted.
2. Invalid and incomplete data is correctly handled.
3. Transformation rules produce the expected output.
4. Valid data is loaded correctly into the warehouse.
5. Fact and dimension tables maintain integrity.
6. Analytical queries return accurate results.
7. Reports display correct information.
8. ETL failures are properly handled and logged.
9. Unauthorized users are prevented from accessing restricted functions.
10. Backup and recovery operations work successfully.
11. The system performs acceptably with the expected data volume.
12. No critical or high-severity unresolved defects remain.

---

14. Conclusion

The test case design for the Data Warehouse Management System provides comprehensive coverage of the major functions and quality requirements of the proposed system. The test cases cover normal, boundary, invalid, and exceptional conditions related to data extraction, validation, transformation, loading, warehouse integrity, querying, reporting, security, performance, backup, and recovery.

Different test design techniques such as Equivalence Partitioning, Boundary Value Analysis, Decision Table Testing, and State Transition Testing are applied to improve test effectiveness and coverage.

Successful execution of these test cases will help ensure that the Data Warehouse Management System is correct, reliable, secure, efficient, and suitable for analytical and business decision-making purposes.